RAG Operations Handbook

1. Index and backup

The Weaviate index is backed up every 6 hours to object storage, and backups are retained for 14 days. A full re-index takes about 12 minutes per 10,000 chunks on the standard node.

2. Hybrid search tuning

Hybrid search blends dense vector similarity with BM25 keyword scores. The default alpha is 0.5. For queries that contain product codes such as SKU-7781 or error codes, lower alpha to 0.25 so exact keyword matches dominate. For long natural-language questions, raise alpha to 0.75.

Scores are combined with relative score fusion, which normalises each result list before blending.

3. Monitoring

Dashboards track p95 retrieval latency, with a target below 300 milliseconds, and total token usage against a budget of 2 million tokens per day.

An alert fires when the daily LLM cost exceeds 25 US dollars or when retrieval hit rate on the golden set drops below 0.8.